

# MATH 3551 A — Abstract Algebra I

## Additional Practice Midterm 1 — Version D

Taylor Dupuy, University of Vermont • Fall 2026

For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

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NAME (please print): \_\_\_\_\_

**Directions.** This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem,  $|H|$  divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

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**About Version D.** Generated with GPT-6 Astra in Codex. This set targets about thirty minutes of work in a fifty-minute period; it has not been timed with students. Coverage: §§0.1–0.3, §§1.1–1.6, §2.1, and §§2.3–2.5. Please report errors, ambiguity, out-of-coverage questions, or timing problems to Professor Dupuy.

**Problem 1.** (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and give a one-line reason: a short proof, calculation, or explicit counterexample. Each item is worth 5 points: 2 for the circle and 3 for the reason.

- (a) **T   F**     If  $f: A \rightarrow B$  and  $g: B \rightarrow C$  are functions and  $g \circ f$  is injective, then  $f$  is injective.
- (b) **T   F**     The relation on  $\mathbb{Z}$  defined by  $a \sim b$  if  $a + b$  is even is an equivalence relation.
- (c) **T   F**     The invertible symmetric  $2 \times 2$  real matrices form a subgroup of  $\mathrm{GL}_2(\mathbb{R})$  under multiplication. (Symmetric means  $A^T = A$ .)
- (d) **T   F**     In  $Q_8$ , with  $ij = k$  and  $j^2 = -1$ , one has  $ij^{-1} = k$ .
- (e) **T   F**     Every nonzero element of the additive group  $(\mathbb{Q}, +)$  has infinite order.
- (f) **T   F**     In every group,  $a^6 = 1$  implies  $a^3 = 1$ .

**Problem 2.** (25 points) **Subgroups of the integers.**

Work in the additive group  $(\mathbb{Z}, +)$ . Write  $d\mathbb{Z} = \{dn : n \in \mathbb{Z}\}$ .

- (a) (8 points) Prove that  $\langle 28, 42 \rangle = 14\mathbb{Z}$  by showing both inclusions.
- (b) (6 points) Determine  $4\mathbb{Z} \cap 6\mathbb{Z}$ , and justify your answer.
- (c) (11 points) List every subgroup  $H$  satisfying  $12\mathbb{Z} \subseteq H \subseteq \mathbb{Z}$ , and draw their subgroup lattice. Put larger subgroups higher, and connect two subgroups when one contains the other and no listed subgroup lies strictly between them. Briefly explain why your list is complete.

**Problem 3.** (25 points) **A rational matrix group.**

For  $t \in \mathbb{Q}$  define

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad H = \{U(t) : t \in \mathbb{Q}\} \subseteq \mathrm{GL}_2(\mathbb{Q}).$$

- (a) (8 points) Compute  $U(s)U(t)$  and  $U(t)^{-1}$ . Use your formulas to prove that  $H$  is a subgroup of  $\mathrm{GL}_2(\mathbb{Q})$ .
- (b) (9 points) Prove that  $\varphi: (\mathbb{Q}, +) \rightarrow H$ ,  $\varphi(t) = U(t)$ , is an isomorphism.
- (c) (8 points) Prove that  $H$  is not cyclic. *Hint:* If a proposed generator is  $U(q)$  with  $q \neq 0$ , consider  $U(q/2)$ .

**Problem 4.** (20 points) **Homomorphisms into  $S_3$ .**

Let  $C = \langle x \rangle$  be a cyclic group of order 6.

- (a) (8 points) Prove that choosing a homomorphism  $f: C \rightarrow S_3$  is equivalent to choosing an element  $y \in S_3$  with  $y^6 = 1$ : explain why  $y = f(x)$  must satisfy this condition, and why  $f(x^m) = y^m$  gives a well-defined homomorphism whenever it does.
- (b) (7 points) List all possible choices for  $f(x)$ , and hence determine exactly how many homomorphisms  $C \rightarrow S_3$  there are.
- (c) (5 points) Can any of these homomorphisms be injective? Justify your answer.